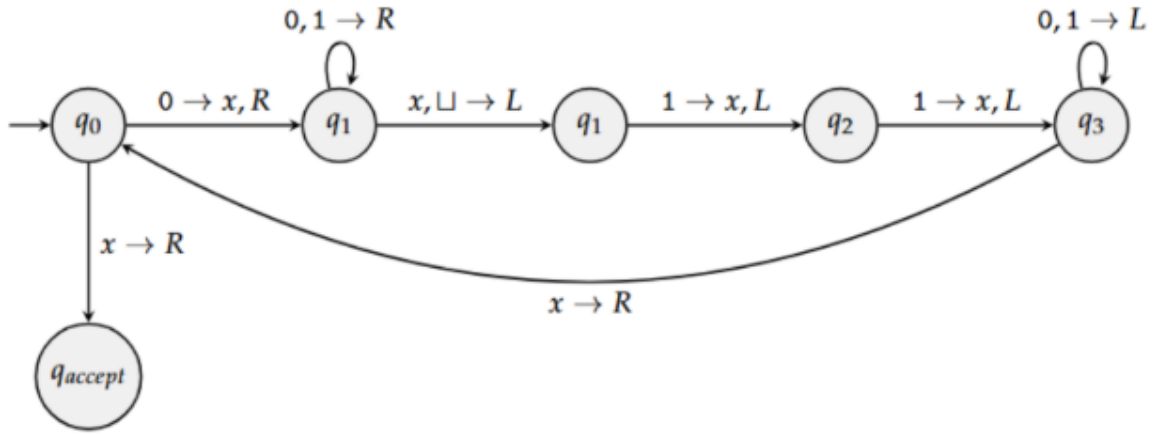


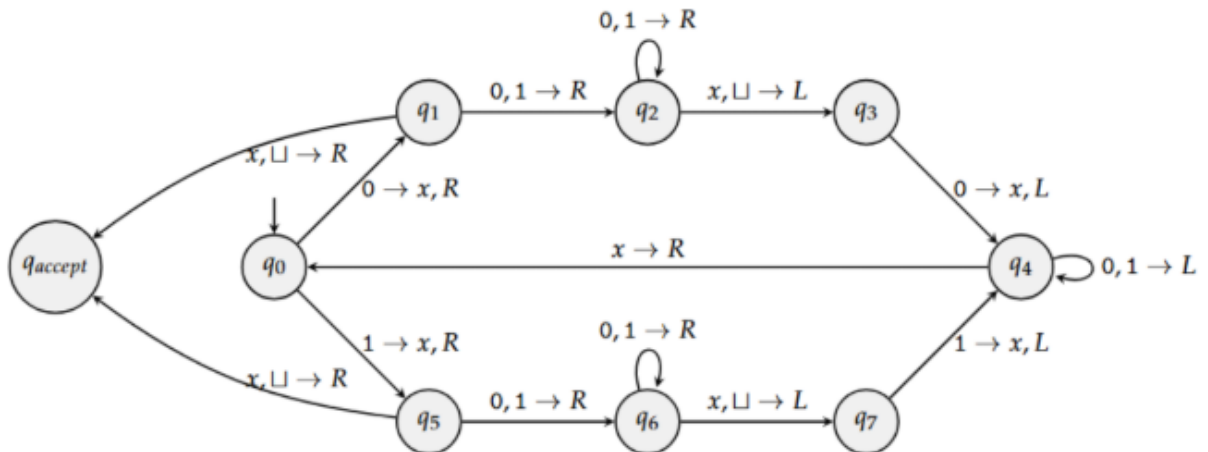
CSE331 : Automata and Computability, Summer 2026		Total Marks: 20	
Name		ID:	Section

Turing Machines

1. Let, $L_1 \rightarrow \{w = 0^n 1^m \mid n, m > 0 \text{ and } 2n = m\}$, where $\Sigma = \{0, 1\}$. Draw the state diagram of a TM that decides L_1 . (6 points)



2. Let, $L_2 \rightarrow \{w \in \Sigma^* \mid w \text{ is an odd length palindrome}\}$, where $\Sigma = \{0, 1\}$. Draw the state diagram of a TM that decides L_2 . (6 points)



3. Write the current state and current head position of a TM in a configuration $01q_5100$. (2 points)

Answer: **Current state:** q_5

- **Current head position:** The head is reading the first 11 after q_5 , which is the **third tape symbol**.

Turing Decidable and Recognizable

Claim: The class of Turing-recognizable languages is closed under concatenation.

Let L_1 and L_2 be recognized by TMs M_1 and M_2 . A student gives the following proof:

Construct TM M . On input w :

1. List every possible split $w = xy$.
2. For each split, run M_1 on x , followed by M_2 on y .
3. If both machines accept for any split, accept.
4. If no split works, reject.

This method works when M_1 and M_2 are **deciders**, but it has a bug when they are **recognizers**. Identify and explain the bug. **[2 marks]**

A recognizer may loop forever on an input that it does not accept. Therefore, if M_1 or M_2 loops for an earlier split, the machine will never examine the remaining splits.

Also, we cannot safely conclude that no split works and reject because some simulations may never halt.

Write the correct proof that Turing-recognizable languages are closed under concatenation. **[4 marks]**

Suppose M_1 recognizes L_1 and M_2 recognizes L_2 . Construct a TM M that recognizes L_1L_2 .

On input w :

1. List all possible splits $w = xy$.
2. Set the step limit $t = 1$.
3. For every split $w = xy$:
 - Run M_1 on x for at most t steps.
 - Run M_2 on y for at most t steps.
4. If both machines accept for any one split within t steps, accept w .
5. Otherwise, increase t by 1 and repeat Step 3.

If $w \in L_1L_2$, some split $w = xy$ has $x \in L_1$ and $y \in L_2$. Both machines will eventually accept, so a sufficiently large step limit will find them.

If $w \notin L_1L_2$, the machine may continue forever, which is allowed for a recognizer.

Therefore, Turing-recognizable languages are closed under concatenation.

Bonus (4 Marks)

Write the proof that Turing recognizable languages are closed under the Kleene star operation.

Suppose M_1 recognizes L . Construct a TM M that recognizes L^* .

On input w :

1. If $w = \epsilon$, accept.
2. List every possible division

$$w = x_1x_2 \cdots x_k,$$

where every x_i is nonempty.

3. Set the step limit $t = 1$.
4. For every possible division, run M_1 on each substring x_i for at most t steps.
5. If every substring in any one division is accepted within t steps, accept w .
6. Otherwise, increase t by 1 and repeat Step 4.

If $w \in L^*$, there is a division in which every substring belongs to L . For a sufficiently large t , all the corresponding simulations will accept.

If $w \notin L^*$, the machine may continue forever.

Therefore, Turing-recognizable languages are closed under the Kleene star operation.